Website Conversion Funnel Analysis

Python + SQL

Identifying conversion leakage, device-specific purchase friction, and the highest-priority areas for investigation

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90,400
users analysed

452
converted users

0.50%
overall conversion

Executive snapshot

The funnel is not suffering from lack of traffic; it is losing users during key purchase transitions.

MOST IMPORTANT NUMBERS

452

completed purchases from 90,400 users

92.50%

abandonment at Payment → Confirmation

39,170

users lost between Search → Payment

4×

Mobile conversion advantage over Desktop

Primary funnel issue

Search → Payment creates the largest absolute leakage; Payment → Confirmation has the highest abandonment rate.

Segment opportunity

Desktop brings 66.6% of users but converts at only 0.25%, compared with 1.00% on Mobile.

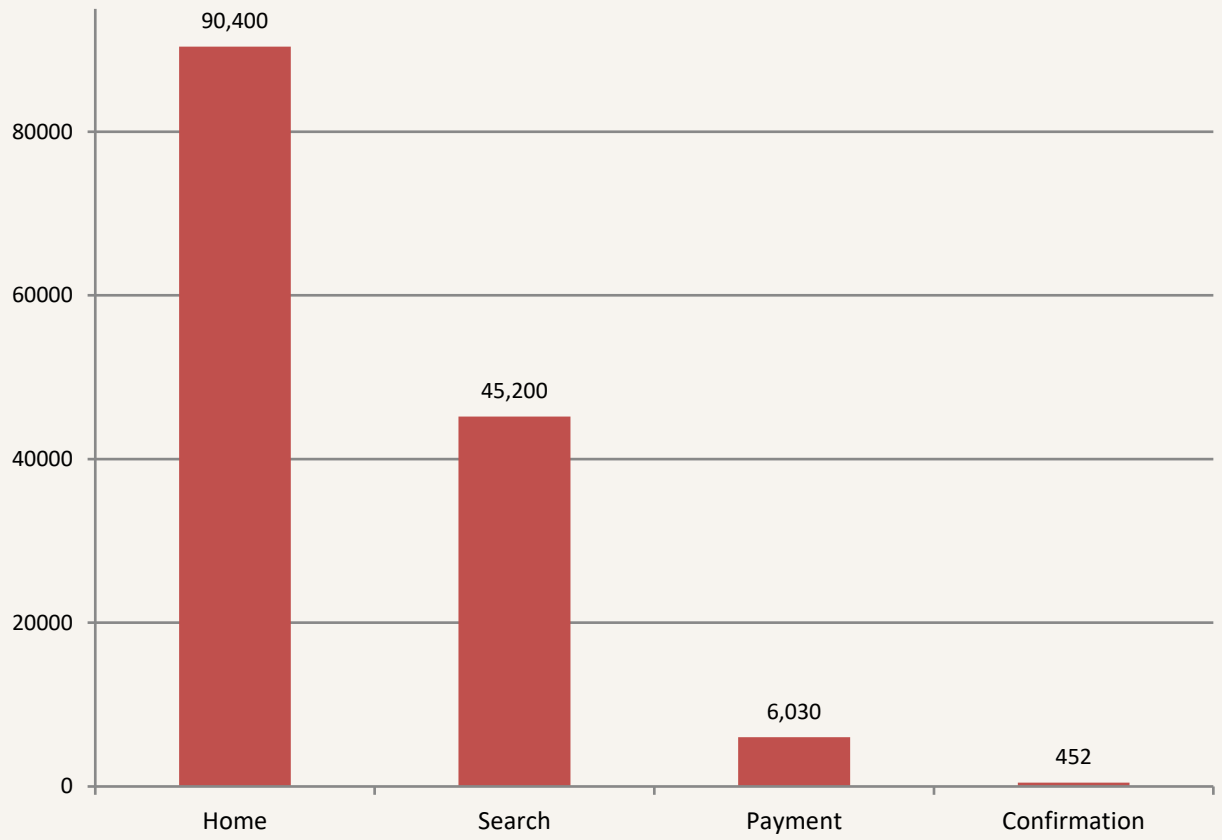
Next decision

Prioritise payment completion and Desktop journey diagnostics before increasing acquisition spend.

Bottom line: the biggest opportunity is converting users already in the funnel, not simply acquiring more traffic.

Where customers are leaving the funnel

Two bottlenecks matter: one by volume, one by proportional abandonment.



Largest absolute loss

Search → Payment lost 39,170 users. This is the biggest volume leak in the purchase journey.

Highest abandonment rate

Payment → Confirmation lost 92.50% of payment users. These users were already near purchase completion.

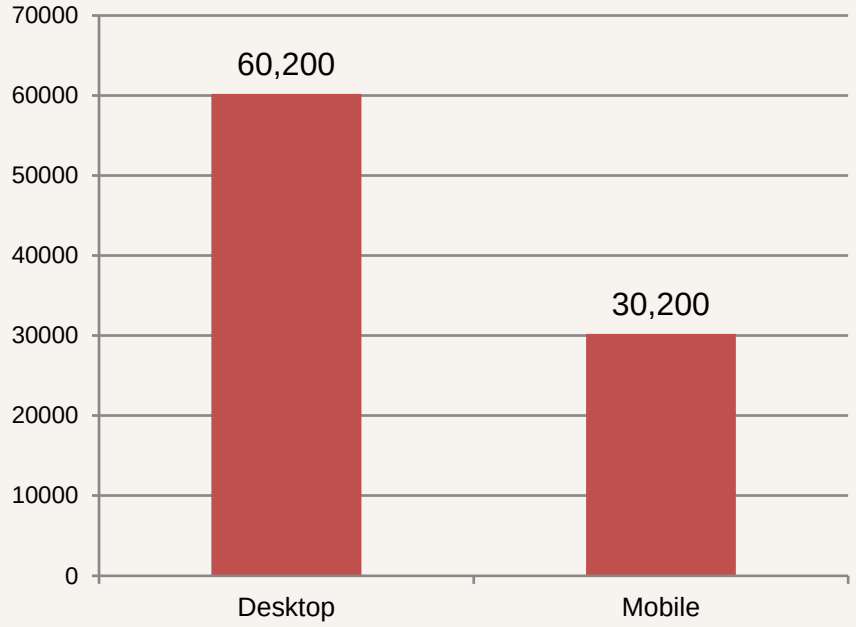
Stakeholder action

Treat Search → Payment and Payment → Confirmation as separate optimisation priorities.

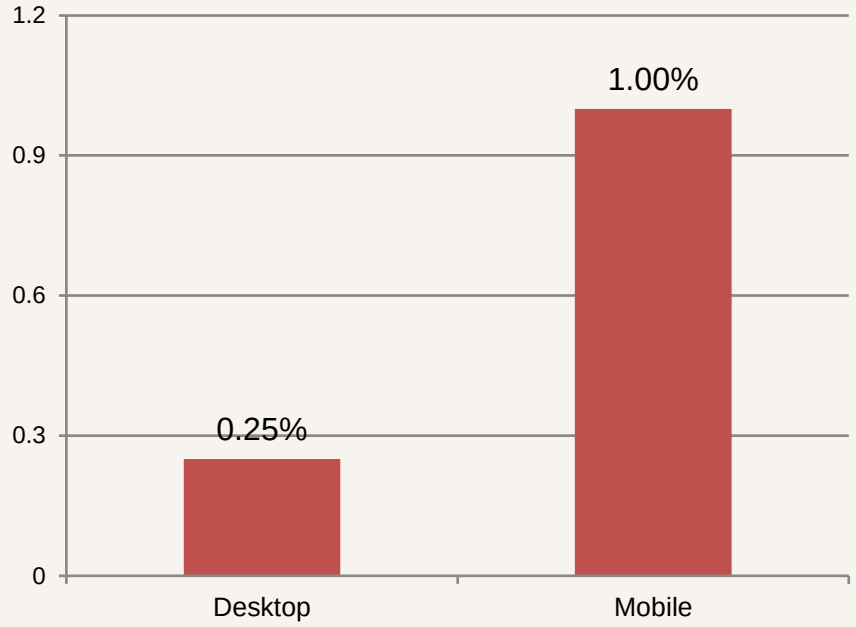
Overall conversion: 452 / 90,400 = 0.50%. Confirmation row has no downstream stage, so post-confirmation conversion is not applicable.

Desktop is the largest untapped segment

Desktop brings most users but converts much worse after initial engagement.



User volume by device



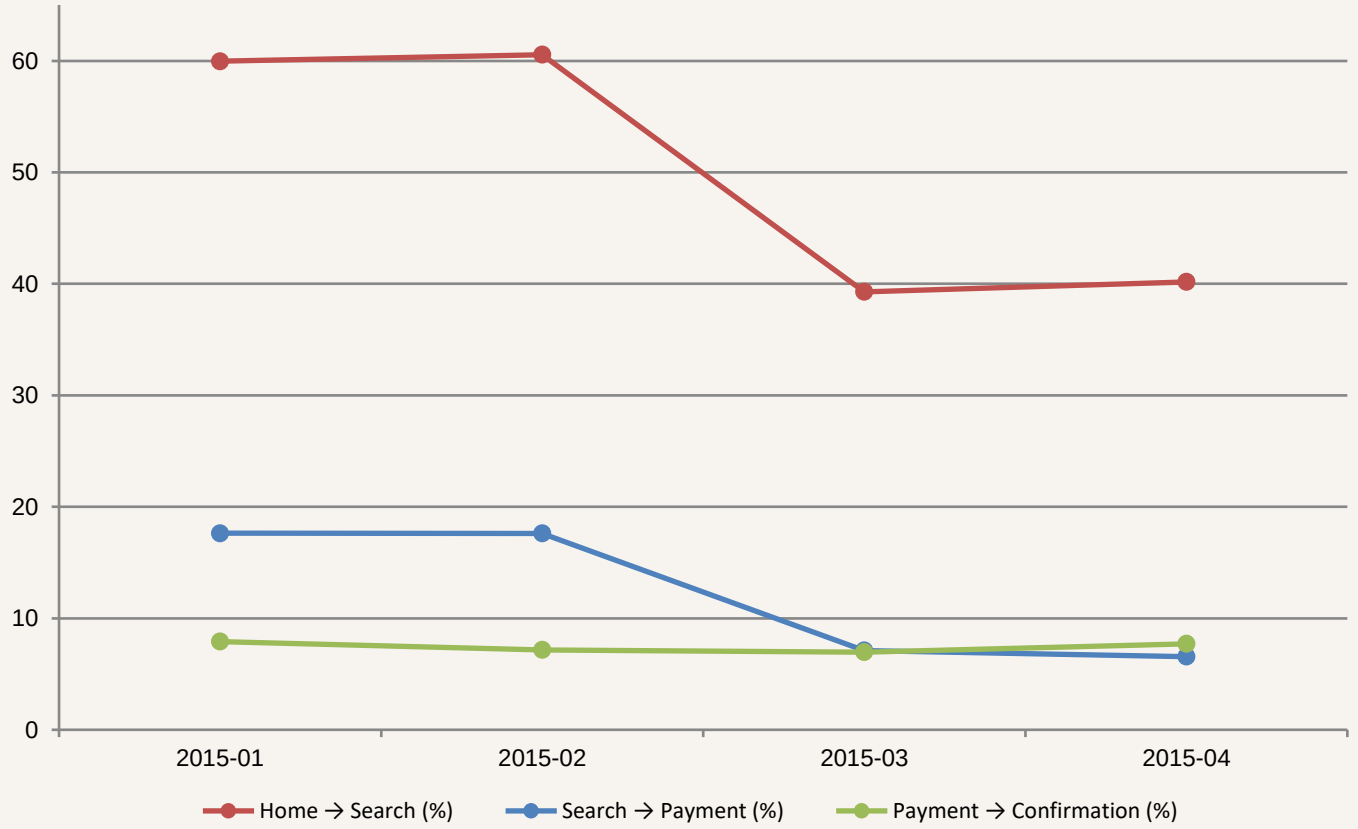
Overall conversion rate

Both devices convert 50.00% from Home → Search, but Desktop falls behind later in the journey.

Implication: Investigate Desktop checkout and purchase-flow friction before spending more on acquisition.

Later monthly user groups perform weaker

March and April groups show lower progression, but the dataset cannot prove a time-based cause.



0.77% → 0.19%

overall conversion fell from Feb group to Mar group

-21.27 pp

Home → Search decline from Feb to Mar

-10.51 pp

Search → Payment decline from Feb to Mar

Note: funnel-stage tables do not contain event timestamps; monthly groups are based on the date recorded in the users table.

Recommended priorities

Focus resources on the bottlenecks most likely to improve conversion from existing traffic.

Priority	Finding	Action
1	**Payment → Confirmation has 92.50% abandonment**	Review payment logs, OTP/gateway failures, checkout complexity, and customer feedback.
2	**Desktop converts 4× worse than Mobile**	Run Desktop usability testing and simplify Search → Payment → Confirmation flow.
3	**Search → Payment loses 39,170 users**	Improve search relevance, product information, trust signals, and offer visibility.

Do not prioritise gender-specific optimisation at this stage as the differences are small compared with stage and device gaps.

Recommendations should be validated before rollout.

What needs to be validated next

The dataset shows where users leave; it does not prove the exact cause.

LIMITATIONS

- **No revenue column:** financial impact of conversion improvements cannot be quantified.
- **No event timestamps in page tables:** monthly analysis uses user-table dates, not exact stage timing.
- **No cause-level data:** payment errors, pricing, campaigns, and customer feedback are unavailable.

NEXT DATA TO COLLECT

Payment failure logs • Session recordings • Page load timings • Traffic source data • Customer feedback • A/B test results

Overall: Prioritize payment completion and Desktop purchase-flow diagnostics before spending more on acquisition.